

Project 1: Customer Relationship Management (CRM) System

1. Abstract

The Customer Relationship Management (CRM) project focuses on developing a centralized platform to manage a company's interactions with its current and potential customers. Developed using the **Agile methodology**, the system is built through iterative cycles or "Sprints," ensuring that stakeholders can provide feedback every 2–4 weeks. The primary goal is to organize, automate, and synchronize sales, marketing, and customer service departments. Key features include lead tracking, contact management, and an analytics dashboard to monitor sales performance.

2. Introduction

- **2.1 Introduction:** CRM is a strategy for managing all your company's relationships and interactions with customers.
- **2.2 Problem Identification:** Many businesses rely on scattered spreadsheets or manual entry, leading to lost leads, double-data entry, and a lack of real-time insights.
- **2.3 Need of the project:** There is a critical need for a cloud-based, automated system that provides a 360-degree view of the customer to improve retention and revenue.
- **2.4 Project Scheduling:** Under Agile, the project is scheduled into four Sprints: Sprint 1 (Requirements & Database), Sprint 2 (Lead Management), Sprint 3 (Dashboard & UI), and Sprint 4 (Testing & Deployment).
- **2.5 Objectives:** To automate lead management, improve data accessibility across teams, and generate automated sales reports.

3. Software Requirement Specification

- **3.1 Purpose:** To provide a unified interface for the sales and support team to track customer life cycles.
- **3.2 Scope:** The project covers Lead Management, Contact Management, Task Tracking, and basic Analytics.
- **3.3 Hardware/Software Requirement:**
 - **Hardware:** Intel i5 Processor, 8GB RAM, 256GB SSD.

- **Software:** Windows 10/11, JDK 17, MySQL 8.0.
- **3.4 Tools:** Eclipse IDE, Selenium WebDriver (for automation testing), Jira (for Agile tracking), and Maven.
- **3.5 Software Process Model: Agile Scrum Model,** allowing for flexibility and rapid delivery of working modules.

4. System Design

- **4.1 Data Dictionary:** A centralized repository defining all data elements such as CustomerID, LeadStatus, and InteractionDate.
- **4.2 ER Diagram:** Defines relationships between entities like Customer, SalesReps, Leads, and Tickets.
- **4.3 DFD:** Illustrates the flow of information from lead entry to conversion and final support.
- **4.4 Class Diagram:** Represents the object-oriented structure of the system, including classes like User, Account, and Opportunity.

5. Implementation

- **5.1 Program Code:** Focuses on Java-based implementation using Spring Boot for backend and Selenium for functional testing of the login and lead forms.
- **5.2 Output Screens:** Snapshots of the Dashboard, Lead Entry Form, and Customer Detail View.

6. Testing

- **6.1 Test Data:** Use of mock customer data, valid/invalid email formats, and boundary values for credit limits.
- **6.2 Test Result:** Success reports from Selenium automated scripts confirming that lead conversion logic works under load.

7. User Manual

- **7.1 How to use project guidelines:** Step-by-step instructions on logging in, creating a lead, and generating a monthly report.
- **7.2 Screen Layouts and Description:** Detailed labels for the navigation bar, search filters, and profile settings.

8. Project Applications and Limitations

- **Applications:** Suitable for real estate, banking, and e-commerce industries to track client queries.
- **Limitations:** Requires high-speed internet for cloud synchronization; limited offline functionality.

9. Conclusion and Future Enhancement

- **Conclusion:** The CRM system effectively solves communication gaps within the team through Agile delivery.
- **Future Enhancement:** Integration of AI-driven chatbots for 24/7 customer support and predictive lead scoring.

10. Bibliography & References

- Referencing official Java documentation and Agile Manifesto principles.